Noregs miljø- og biovitskaplege universitet

Institutt for datavitskap

Digitalisering på Ås

INF230
Datahandtering og -analyse

1 Ontologiar

Noregs miljø- og biovitskaplege universitet

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1 Ontologiar

1.2 Vevontologispråket OWL

INF230

3. september 2026

Termliste

nn **blank node** *m.*

en **blank node**

nn **DIKV-pyramide** *m.*

en **DIKV pyramid**

nn **informasjon** *m.*

en **information**

nn **kunnskapsbase** *m.*

en **knowledge base**

nn **ontologi** *m.*

en **ontology**

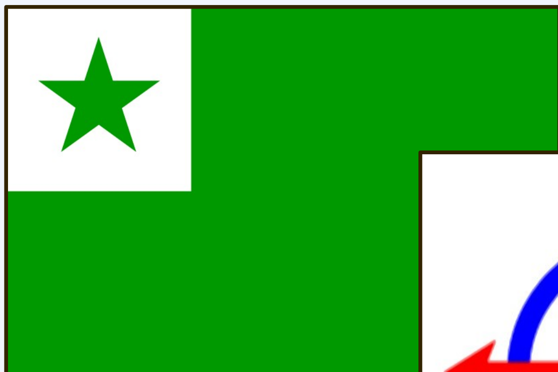
nn **semantisk artefakt** *n.*

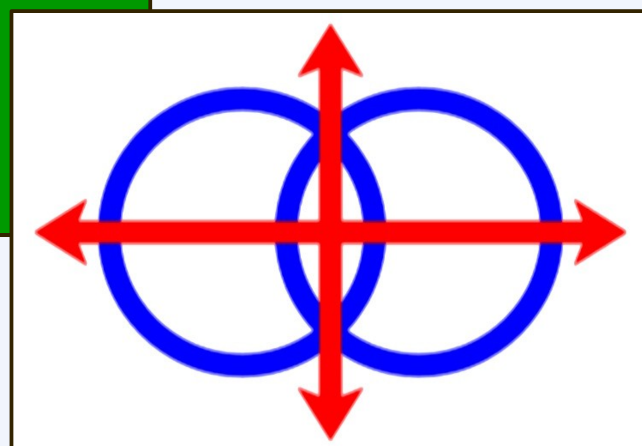
en **semantic artefact**

Why OWL?

„One World Language"

or now, "Web Ontology Lanugage (OWL)," based on RDF Schema (RDFS)





Protégé is an important tool for OWL, highly recommended:

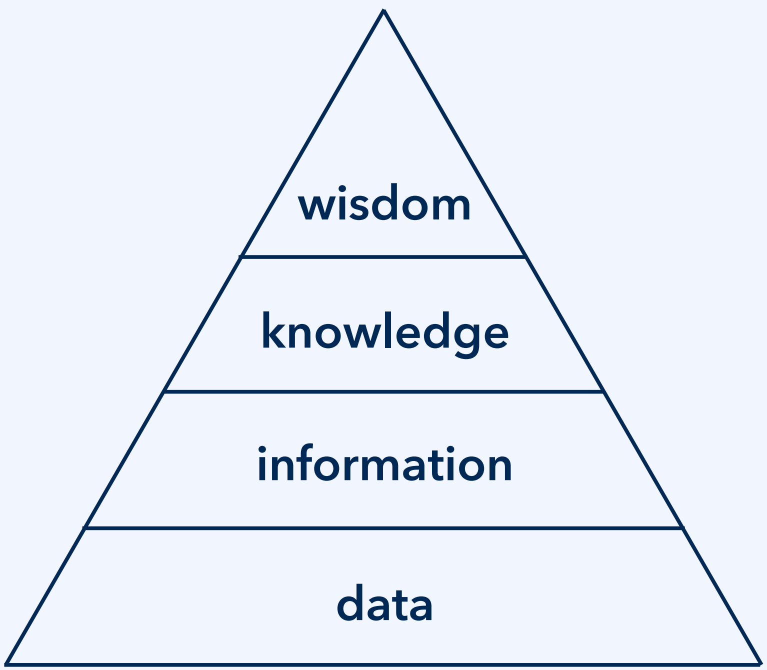
https://protege.stanford.edu/

Why OWL?

Hierarchy of data, information, knowledge, and wisdom (**DIKW pyramid**[1,2]).

[1] J. Rowley, *J. Inform. Sys.* **33**: 163–180, doi:10.1177/0165551506070706, **2009**.

[2] B. Schembera, in *Proc. DCLXVI 2024*, pp. 122–132, Springer, doi:10.1007/978-3-031-89274-5_9, **2025**.



Pragmatic competency and interoperability, including agreed good practices.

Epistemic metadata documentation: Establish the knowledge status.

Semantic interoperability: Data become information if their meaning is agreed.

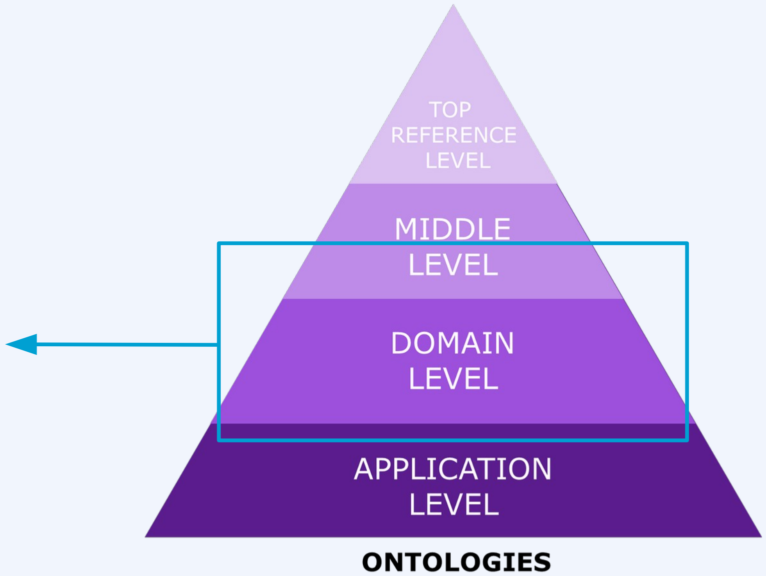
Syntactic interoperability: Data exchanged in an agreed format.

Semantic interoperability

OntoCommons CSA collected and supported the design and alignment of **domain-level interoperability standards**.

The overall analysis of modes of interoperability, relevant tools and components, and recommendations was delivered in the form of **RoDI: The Review of Domain Interoperability**.[1]

The RoDI document also distinguishes **syntactic**, **semantic**, and **pragmatic** interoperability.[1]



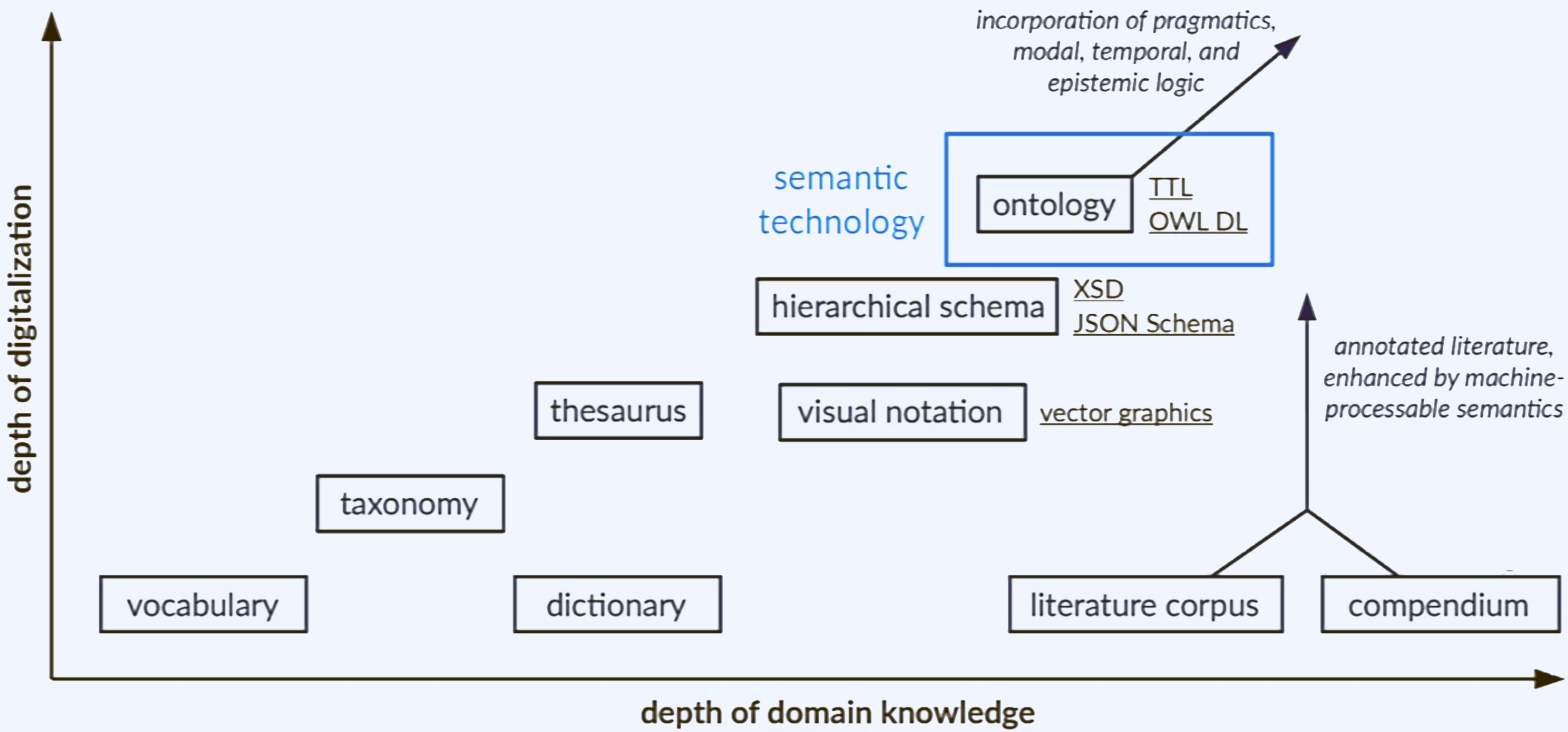
	data	human	organization	software
data (d)	$d \leftrightarrow d$	$d \leftrightarrow h$	$d \leftrightarrow o$	$d \leftrightarrow s$
human (h)	—	$h \leftrightarrow h$	$h \leftrightarrow o$	$h \leftrightarrow s$
organization (o)	—	—	$o \leftrightarrow o$	$o \leftrightarrow s$
software (s)	—	—	—	$s \leftrightarrow s$

Matrix structure for interoperability requirements: Who interoperates with whom?

[1] S. Chiacchiera *et al.*, OntoCommons deliverable 3.8, "Finalized Review of Domain Interoperability," **2023**.

Semantic interoperability

Types of metadata standards (also called **semantic artefacts**):



Making use of ontologies

schema.org

Custom Search

Home Schemas Documentation

Hotel

Thing > Organization > LocalBusiness > LodgingBusiness > Hotel
Thing > Place > LocalBusiness > LodgingBusiness > Hotel

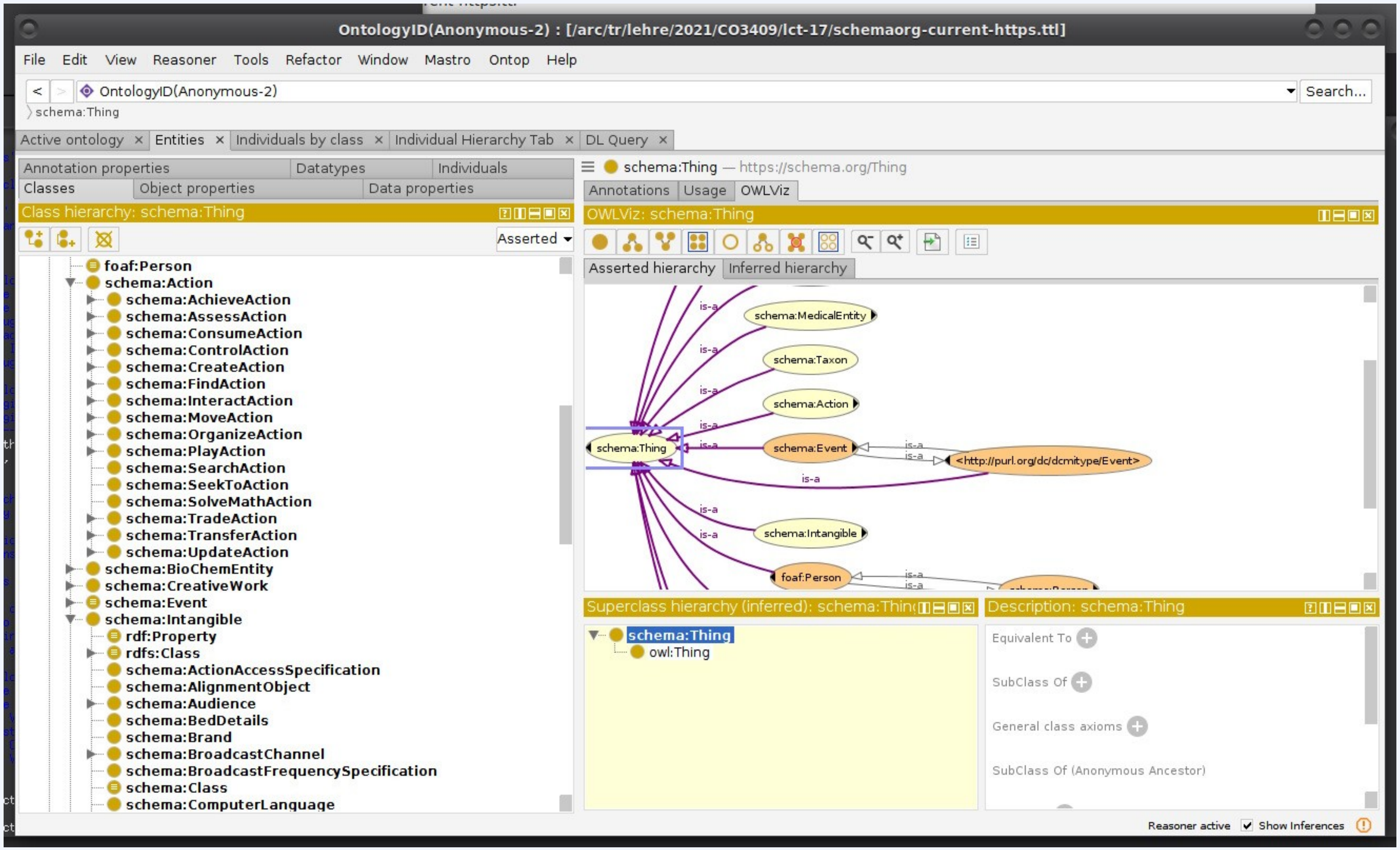
A hotel is an establishment that provides lodging paid on a short-term basis (Source Wikipedia, the free encyclopedia, see http //en.wikipedia.org /wiki/Hotel).
See also the dedicated document on the use of schema.org for marking up hotels and other forms of accommodations.

Property	Expected Type	Description
Properties from LodgingBusiness		
amenityFeature	LocationFeatureSpecification	An amenity feature (e.g. a characteristic or service) of the Accommodation. This generic property does not make a statement about whether the feature is included in an offer for the main accommodation or available at extra costs.
audience	Audience	An intended audience, i.e. a group for whom something was created. Supersedes serviceAudience.

Ferienhof Hotel Garni Oblasser


```
{
  "@context": "http://schema.org",
  "@type": [
    "Hotel"
  ],
  "url": [
    "http://www.ferienhof-oblasser.at",
    "http://maps.mayrhofen.at/?foreignResource=C5EC2DF7-5E30-42AB-9D58-
5607FB343ECA"
  ],
  "address": {
    "@type": "PostalAddress",
    "name": "Oblasser, Ferienhof",
    "streetAddress": "Hochstegen 835",
    "addressLocality": "Mayrhofen",
    "postalCode": "6290",
    "addressCountry": "AT",
    "telephone": "0043 5285 64666",
    "faxNumber": "0043 5285 64559",
    "email": "info@ferienhof-oblasser.at",
    "url": "http://www.ferienhof-oblasser.at"
  },
  "name": "Ferienhof Hotel Garni Oblasser",
  "description": [
    "Children up to the age of 6 go free, up to the age of 12 a 50%
price reduction.\nIncludes: Heating, garage, water, electricity,
local taxes and bed linen.\nExcludes: final cleaning charge",
```

Making use of ontologies



[1]Dokumention schema.org: https://schema.org/docs/full.html.

[2]Ontology in TTL-format: https://schema.org/version/latest/schemaorg-current-https.ttl.

Resource description framework schemas

The basic construct for specifying a set in RDFS is called an `rdfs:Class`. Since RDFS is expressed in RDF, the way we express that something is a class is with a triple—in particular, a triple in which the predicate is `rdf:type`, and the object is `rdfs:Class`. Here are some examples that we will use in the following discussion:

```
:AllStarPlayer rdf:type rdfs:Class .
:MajorLeaguePlayer rdf:type rdfs:Class .
:Surgeon rdf:type rdfs:Class .
:Staff rdf:type rdfs:Class .
:Physician rdf:type rdfs:Class .
```

These are triples in RDF just like any other; the only way we know that they refer to the schema rather than the data is because of the use of the term in the `rdfs:` namespace, `rdfs:Class`.

whenever we introduce a new RDFS resource, we will answer the question "What does it mean?" with an answer of the form "In these circumstances (defined by some pattern of triples), you can add (infer) the following new triples." We already saw how to do this with the type propagation rule for `rdfs:subClassOf`; now we will demonstrate this principle using another of the most fundamental terms in RDFS: `rdfs:subPropertyOf`.

Allemang *et al.*, Section 8.2: The RDF schema language

Resource description framework schemas

There are a number of relationships that can hold between a person and the firm; we can call them `contractsTo`, `freeLancesTo`, `indirectlyContractsTo`, `isEmployedBy`, and `worksFor`.

If we assert any of these statements about some person, then we would like to infer that that person `worksFor` the firm.

All these relationships can be expressed in RDFS using the `rdfs:subPropertyOf` relation:

```
:freeLancesTo rdfs:subPropertyOf contractsTo .
:indirectlyContractsTo rdfs:subPropertyOf contractsTo .
:isEmployedBy rdfs:subPropertyOf worksFor .
:contractsTo rdfs:subPropertyOf worksFor .
```

Sidebar 8.1 OO comparison

The construct `rdfs:subPropertyOf` has no direct analog in OOP, where properties are not first-class entities (i.e., they cannot be related to one another, independent of the class in which they are defined). For this reason, unlike the case of `rdfs:subClassOf`, OO programmers have no conflict with a similar known concept. The only source of confusion is that subproperty diagrams like the preceding one are sometimes mistaken for class diagrams.

Allemang *et al.*, Section 8.2: The RDF schema language

Resource description framework schemas

In RDFS, the properties `rdfs:domain` and `rdfs:range` have meanings inspired by the mathematical uses of these words. A property p can have an `rdfs:domain` and/or an `rdfs:range`. These are specified, as is everything in RDF, via triples:

```
:p rdfs:domain :D.
:p rdfs:range :R.
```

The informal interpretation of this is that the relation `p` relates values from the class `D` to values from the class `R`. `D` and `R` need not be disjoint, or even distinct.

In RDFS, domain and range give some information about how the property P is to be used; domain refers to the subject of any triple that uses P as its predicate, and range refers to the object of any such triple. When we assert that property P has domain D (respectively, range R), we are saying that whenever we use the property P, we can infer that the subject (respectively, object) of that triple is a member of the class D (respectively, R). In short, domain and range tell us how P is to be used.

Allemang *et al.*, Section 8.2: The RDF schema language

Writing up an RDF schema

animals:Fox a owl:Class.

concepts are of the type owl:Class

taxonomy specified using rdfs:subClassOf

animals:Fox rdfs:subClassOf animals:Canidae.
animals:Canidae rdfs:subClassOf animals:Mammalia.

animals:isNaturalEnemyOf a owl:ObjectProperty.

relations are of the type owl:ObjectProperty

animals:isNaturalEnemyOf rdfs:domain animals:Predator.

rdfs:domain and similarly **rdfs:range**

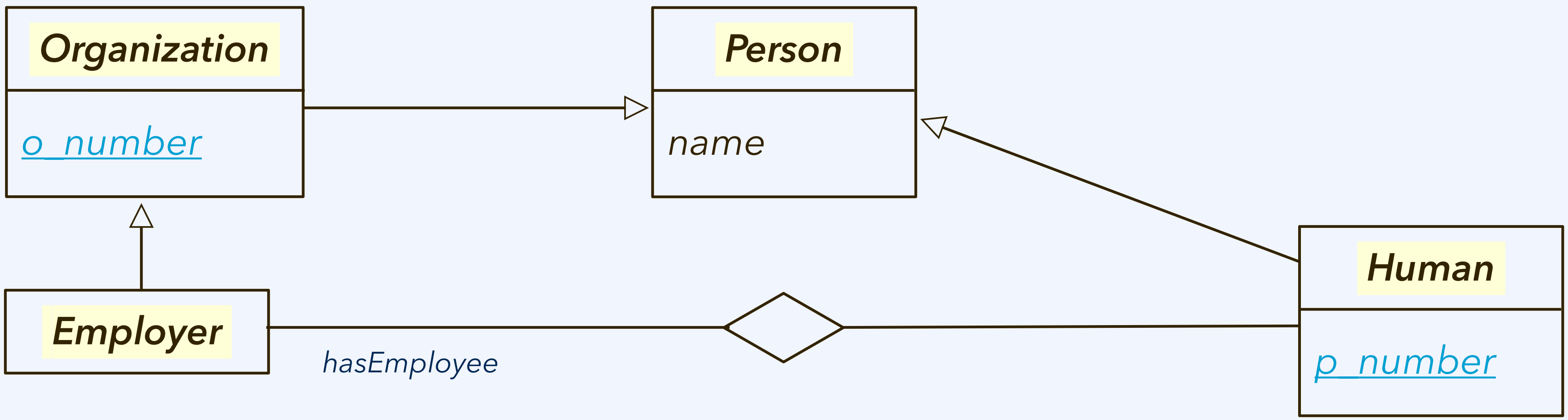
uni:hasSubmissionTimestamp a owl:DatatypeProperty.

uni:hasSubmissionTimestamp rdfs:range xs:dateTime;
rdfs:subPropertyOf uni:hasTimestamp.

elementary **datatype properties**

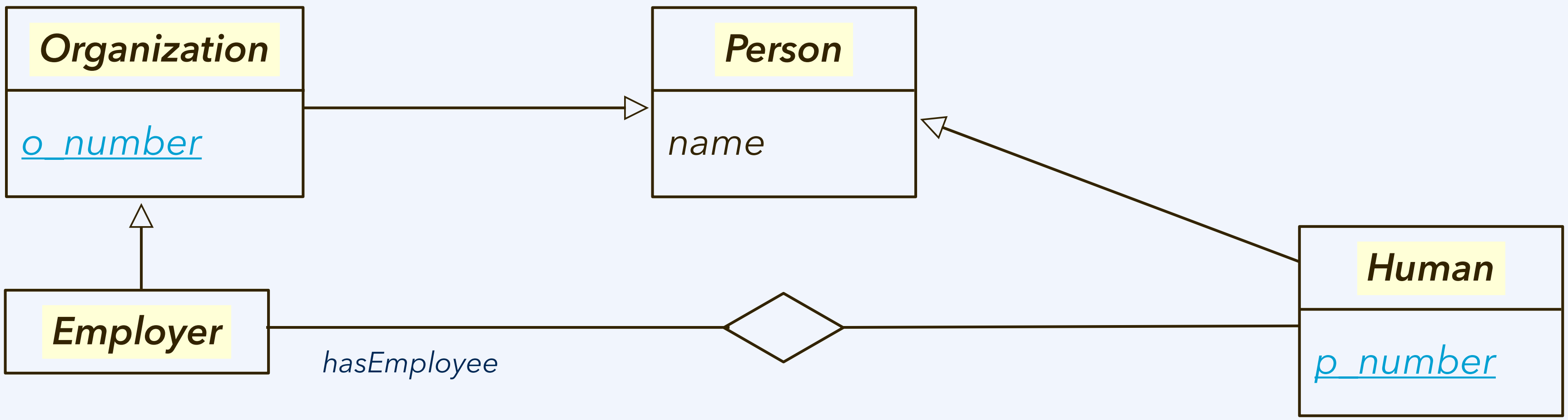
hierarchy for datatype properties and **relations** (object propertes)

Can we write this ER diagram in RDFS?



- Organization is a subclass of Person.
    - *Or: Every Organization is a Person.*
- Employer is a subclass of Organization. Human is a subclass of Person.
- There is a relationship type called hasEmployee.
- The domain of hasEmployee is Employer, the range is Human.
    - *Or: Every subject of hasEmployee is an Employer, every object a Human.*
- Some objects can have elementary data properties, *e.g.*, p_number.
    - *Every individual that has a p_number is a Human.*

Can we write this ER diagram in RDFS?



taxonomy specified
using rdfs:subClassOf

animals:Fox a owl:Class.

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rdfs:domain and similarly **rdfs:range**

uni:hasSubmissionTimestamp a owl:DatatypeProperty.

elementary **datatype**
properties

Knowledge bases: TBox and ABox

A **knowledge base** for linked data consists of two components:

> **Definition:** A **knowledge base**, given by $K = (T, A)$, consists of an ontology T, describing universals, and a set of assertions A describing concrete instances of these universals.

ABox = knowledge graph

particular:	entity **individual** object	**relationship**	(sometimes: attribute) **property**
universal:	entity type **concept** class	relationship type **relation** (in OWL: ObjectProperty)	(sometimes: attribute type) **attribute** (in OWL: DatatypeProperty)

TBox = ontology

Knowledge bases: TBox and ABox

1.2 What is an ontology?

The most quoted (but problematic!) definition is the following one by Tom Gruber:

Definition 1.1 ([Gru93]). *An ontology is a specification of a conceptualization.*

You may see this quote especially in older scientific literature on ontologies, but it has been superseded by other, more precise ones, for Gruber's definition is unsatisfactory for several reasons: what is a "conceptualization" exactly, and what is a "specification"? Using two nebulous terms to describe a third one

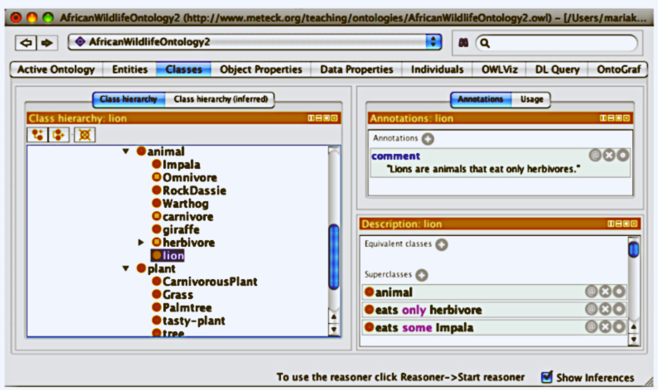

Figure 1.2: Screenshot of the lion eating only herbivores and at least some impala in the Protégé ontology development environment.

Definition 1.2 ([SBF98]). *An ontology is a formal, explicit specification of a shared conceptualization.*

However, this still leaves us with the questions as to what a "conceptualization" is and what a "formal, explicit specification" is, and why and how "shared"? Is it shared enough when, say, you and I agree on the knowledge represented in the ontology, or do we need a third one or a whole group to support it? A comprehensive definition is given in Guarino's landmark paper on ontologies [Gua98] (revisited in [GOS09]):

Definition 1.3 ([Gua98]). *An ontology is a logical theory accounting for the intended meaning of a formal vocabulary, i.e. its* ontological commitment *to a particular* conceptualization *of the world. The intended models of a logical language using such a vocabulary are constrained by its ontological commitment. An ontology indirectly reflects this commitment (and the underlying conceptualization) by approximating these intended models.*

A broader scope is also described in [Gua09], and a more recent overview about definitions of "an ontology" versus Ontology in philosophy can be found in [GOS09], which refines in a step-wise and more precise fashion Definitions 1.2 and 1.3. It is still not free of debate [Neu17], though, and it is a bit of a mouthful as definition. A simpler definition is given by the developers of the World Wide Web Consortium's standardised ontology language OWL[6]:

Definition 1.4 ([HPSvH03]). *An ontology being equivalent to a Description Logic knowledge base.*

That last definition has a different issue, and is unduly restrictive, because 1) it surely is possible to have an ontology that is represented in another logic language

FOL semantics for RDF schema

model theory of RDFS using first-order logic (FOL)

includes a ternary predicate T –to represent RDF triples – and two unary predicates C and P that will represent the membership of individuals to "`rdfs:Class`" and "`rdf:Property`", respectively.

The basic axioms primitively define `subClass`, `subProperty`, `domain`, `range` in terms of `type` in the obvious way –as in set theory[1]:

$$\forall a, b\ (a, \mathtt{sc}, b) \longrightarrow C(a) \wedge C(b) \wedge \forall x\ (x, \mathtt{type}, a) \to (x, \mathtt{type}, b) \tag{1}$$

$$\forall a, b\ (a, \mathtt{sp}, b) \longrightarrow P(a) \wedge P(b) \wedge \forall x, y\ (x, a, y) \to (x, b, y) \tag{2}$$

$$\forall a, c\ (a, \mathtt{dom}, c) \longrightarrow \forall x, y\ (x, a, y) \to (x, \mathtt{type}, c) \tag{3}$$

$$\forall a, d\ (a, \mathtt{range}, d) \longrightarrow \forall x, y\ (x, a, y) \to (y, \mathtt{type}, d) \tag{4}$$

To cope with reflexivity and transitivity of the subclass and subproperty relations we have also the following axioms:

$$\forall a, b, c\ (a, \mathtt{sc}, b) \wedge (b, \mathtt{sc}, c) \longrightarrow (a, \mathtt{sc}, c) \tag{5}$$

$$\forall a\ C(a) \longrightarrow (a, \mathtt{sc}, a) \tag{6}$$

$$\forall a, b, c\ (a, \mathtt{sp}, b) \wedge (b, \mathtt{sp}, c) \longrightarrow (a, \mathtt{sp}, c) \tag{7}$$

$$\forall a P(a) \longrightarrow (a, \mathtt{sp}, a) \tag{8}$$

E. Francon *et al.*, «The logic of extensional RDFS», in *Proc. ISWC 2013* (Springer, LNCS **8218**) 101–116, doi:10.1007/978-3-642-41335-3_7, **2013**

FOL semantics for RDF schema

$$\forall a, b\ (a, \texttt{dom}, b) \longrightarrow P(a) \wedge C(b) \tag{9}$$

$$\forall a, b\ (a, \texttt{range}, b) \longrightarrow P(a) \wedge C(b) \tag{10}$$

$$\forall a, b\ (a, \texttt{type}, b) \longrightarrow C(b) \tag{11}$$

$$\forall a, b, c\ (a, b, c) \longrightarrow P(b) \tag{12}$$

$$P(\texttt{sc}) \wedge P(\texttt{sp}) \wedge P(\texttt{dom}) \wedge P(\texttt{range}) \wedge P(\texttt{type}) \tag{13}$$

The above axioms define the semantics for the `subClass`, `subProperty`, `domain` and `range` predicates.

The W3C specification [7] introduces in a "non-normative"status an *extensional* version of RDFS, in which `subClass`, `subProperty`, `domain`, `range` are defined precisely as having the usual set theoretical meaning. This is achieved by

$$\forall a, b\ (a, \texttt{sc}, b) \longleftrightarrow C(a) \wedge C(b) \wedge \forall x\ (x, \texttt{type}, a) \rightarrow (x, \texttt{type}, b) \tag{14}$$

$$\forall a, b\ (a, \texttt{sp}, b) \longleftrightarrow P(a) \wedge P(b) \wedge \forall x, y\ (x, a, y) \rightarrow (x, b, y) \tag{15}$$

$$\forall a, c\ (a, \texttt{dom}, c) \longleftrightarrow \forall x, y\ (x, a, y) \rightarrow (x, \texttt{type}, c) \tag{16}$$

$$\forall a, d\ (a, \texttt{range}, d) \longleftrightarrow \forall x, y\ (x, a, y) \rightarrow (y, \texttt{type}, d) \tag{17}$$

E. Francon *et al.*, «The logic of extensional RDFS», in *Proc. ISWC 2013* (Springer, LNCS **8218**) 101–116, doi:10.1007/978-3-642-41335-3_7, **2013**

Blank nodes in TTL notation

Sometimes we know that something exists, and we even know some things about it, but we don't know its identity. For instance, suppose we want to represent the conjecture that Shakespeare had a mistress, whose identity remains unknown. But we know a few things about her; she was a woman, she lived in England, and she was the inspiration for *Sonnet 78*.

```
lit:Mistress1 rdf:type bio:Woman;
bio:LivedIn geo:England.
lit:Sonnet78 lit:hasInspiration lit:Mistress1.
```

But if we don't want to have an identifier for the mistress, how can we proceed? RDF allows for a *blank node*, or *bnode* for short, for such a situation.

Turtle instead includes a compact and unambiguous notation for describing blank nodes. A blank node is indicated by putting all the triples of which it is a subject between square brackets ([and]), so:

```
[ rdf:type bio:Woman;
  bio:livedIn geo:England ]
```

Allemang *et al.*, avsnitt 3.9: Blank nodes

Blank nodes in TTL notation

We can refer to this blank node in other triples by including the entire bracketed sequence in place of the blank node. Furthermore, the abbreviation of a for rdf:type is particularly useful in this context. Thus, our entire statement about the mistress who inspired "*Sonnet 78*" looks as follows in Turtle:

```
lit:Sonnet78 lit:hasInspiration [a :Woman;
             bio:livedIn geo:England].
```

This expression of RDF can be read almost directly as plain English: that is, "Sonnet78 has [as] inspiration a Woman [who] lived in England." The identity of the woman is indeterminate.

Allemang *et al.*, avsnitt 3.9: Blank nodes

Existential and universal restrictions

A restriction is a special kind of a class, so it has individual members just like any class. Membership in a restriction class must satisfy the conditions specified by the kind of restriction (owl:allValuesFrom, owl:someValuesFrom, or owl:hasValue), as well as the onProperty specification.

owl:someValuesFrom

owl:someValuesFrom is used to produce a restriction of the form "All individuals for which at least one value of the property P comes from class C." In other words, one could define the class AllStarPlayer as "All individuals for which at least one value of the property playsFor comes from the class AllStarTeam." This is what the restriction looks like:

```
[ a owl:Restriction;
  owl:onProperty :playsFor;
  owl:someValuesFrom :AllStarTeam ]
```

Allemang *et al.*, Section 12.1: Basic OWL: Restrictions

Existential and universal restrictions

An answered question is one that has some value from the class `Answer` for the property `hasSelectedOption`. This can be defined as follows:

```
q:AnsweredQuestion owl:equivalentClass
      [ a owl:Restriction ;
        owl:onProperty q:hasSelectedOption ;
        owl:someValuesFrom q:Answer ] .
```

Since

```
d:WhatProblem q:hasSelectedOption d:STV.
```

and

```
d:STV a q:Answer.
```

are asserted triples, the individual `d:WhatProblem` satisfies the conditions defined by the restriction class.

Allemang *et al.*, Section 12.1: Basic OWL: Restrictions

Existential and universal restrictions

if My Favorite All Star Team is restricted to all players being star players, then every player on my favorite all star team is a star player. Furthermore, if Kaneda and Gonzales play on my favorite all star team, then both of them must be star players. This intuition is shown in triples below (with the usual convention that inferred triples are marked with an asterisk [*]):

```
MyFavoriteAllStarTeam a BaseballTeam ;
                      a [ owl:Restriction ;
                              owl:onProperty :hasPlayer ;
                              owl:allValuesFrom :
                                 StarPlayer ] ;
                      :hasPlayer :Kaneda , :Gonzales .
* :Gonzales a :StarPlayer .
* :Kaneda a :StarPlayer .
```

Allemang *et al.*, Section 12.1: Basic OWL: Restrictions

Glossar

DIKV-pyramide *(en. DIKW pyramid)*

Forklaring:

- **data**: digitalt innhald som blir utveksla i samsvar med ein formell **syntaks**
- **informasjon**: data med veldefinert betydning (dvs. formell **semantikk**)
- **kunnskap**: informasjon brukaren veit er sann eller, i utvida betydning, informasjon med karakterisert **epistemisk status**
- **visdom**: kunnskap som brukaren kan handtere godt i praksis, dvs. med ein **pragmatisk** kontekst som er kjent for brukeren.

Men det er mange forskjellige definisjonar på dette.[1]

[1] J. Rowley, *Journal of Information Systems* **33**(2): 163–180, doi:10.1177/0165551506070706, **2007**.

Glossar

informasjon

Definisjon: utsegn eller mengde av utsegner

- Eller noko som kan lagast om til utsegner, t.d. data med veldefinert semantikk.

kunnskapsbase

Definisjon: par $K = (A, T)$, der A er ein ABoks (kunnskapsgraf) og T er ein TBoks (ontologi)

ontologi

Definisjon: «formell oversikt over eit kunnskapsfelt, representert ved omgrep og relasjonar mellom dei» (M. K. Ådland i SNL)

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Diskusjonstime

Nettskjema

Link to language questionnaire:

https://nettskjema.no/a/inf230lang

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Diskusjonstime

Group formation finalization

INF230

3. september 2026

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INF230
Datahandtering og -analyse

1 Ontologiar

1.1	Den semantiske veven	1.4	Ontologiutvikling
1.2	**Vevontologispråket OWL**	1.5	DOLCE, EMMO og BFO
1.3	Terminologiutvikling	1.6	Semantisk heterogenitet